Notes: Caballero and Engel (Econometrica, 1999)

Explaining Investment Dynamics in U.S. Manufacturing: A Generalized (S, s) Approach

Contents

1 One-sentence summary

This note summarizes Caballero and Engel's paper, which builds and estimates a generalized (S, s) model of lumpy plant-level investment with stochastic fixed adjustment costs, uses aggregate two-digit U.S. manufacturing equipment and structures investment data from 1948–1992 to infer smooth adjustment hazards and nonlinear aggregate dynamics, and shows that the nonlinear model fits and forecasts sectoral investment substantially better than standard linear partial-adjustment or AR models.

2 Research question and motivation

The paper asks how aggregate investment dynamics can be explained when microeconomic investment is intermittent and lumpy rather than smooth. The central empirical puzzle is that plant-level investment data show large, infrequent investment spikes, while standard aggregate investment equations often impose smooth adjustment or linear partial adjustment. Caballero and Engel therefore ask whether an aggregate investment model that is explicitly built from lumpy micro-level adjustment can improve both the economic interpretation and the time-series performance of investment equations.

The motivation has three layers. First, plant-level investment behavior is highly lumpy: large establishments often concentrate a substantial share of their multi-year investment in one year or project. Second, if micro units are not continuously adjusting, aggregate investment should depend not only on the average capital gap but also on the cross-sectional distribution of firm-level disequilibria. Third, because the cross-sectional distribution evolves with past shocks and past adjustment, aggregate investment should display nonlinear dynamics: after accumulated shocks, many firms may be near their adjustment thresholds, making aggregate investment unusually responsive.

3 Main contribution and incremental contribution

3.1 Main contribution

The paper develops a tractable aggregate investment model that starts from lumpy microeconomic adjustment and produces a nonlinear aggregate time-series process. Instead of imposing deterministic (S, s) bands, the model allows fixed adjustment costs to be random across firms and over time. This generates a smooth *adjustment hazard*: the probability that a firm adjusts rises with the absolute size of its capital imbalance. Aggregating this hazard over the cross-sectional density of firm-level imbalances yields an aggregate investment equation whose dynamics depend on the entire distribution of imbalances, not only on the mean gap.

3.2 Incremental contribution relative to existing investment models

The incremental contribution is that the paper bridges three literatures that were previously less integrated: micro models of lumpy/nonconvex adjustment, aggregation with heterogeneous units, and empirical aggregate investment dynamics. Compared with standard partial-adjustment or quadratic adjustment-cost models, the paper provides a micro-founded source of aggregate nonlinearities. Compared with sharp deterministic (S, s) models, the paper introduces a probabilistic hazard that is smoother, easier to aggregate, and flexible enough to nest both sharp (S, s) behavior and approximately linear partial adjustment. Compared with earlier empirical (S, s) applications, the paper estimates the nonlinear aggregate model using only aggregate sectoral investment series and a distributional assumption about aggregate shocks, rather than requiring a separately observed aggregate driving force.

3.3 Why the contribution matters

The key insight is that the state of aggregate investment is not summarized by current fundamentals alone. Because firms adjust only intermittently, past shocks leave a distributional footprint: a sector can have many firms close to adjustment, or many firms far from adjustment. Therefore, the same aggregate shock can generate very different aggregate investment responses depending on the inherited cross-sectional distribution. This state dependence is the paper's core economic mechanism and is what standard linear models miss.

4 Economic environment and model primitives

4.1 Firms and frictionless capital

The economy is a sector composed of many monopolistically competitive firms. Each firm produces with capital and labor and faces a profit function that can be reduced to a function of its capital stock K and a composite shock θ. The frictionless capital stock K^* is the stock that would maximize current profits absent adjustment costs. The central state variable is the log capital imbalance

$$z \equiv \ln(K/K^*),$$

so that $z > 0$ means the firm has excess capital relative to frictionless desired capital, and $z < 0$ means the firm has a capital shortage.

The profit function can be written as

$$\Pi(z, K^*) = \pi(z)K^*,$$

where $\pi(z)$ is maximized near the target point and falls when the firm is too far above or below its frictionless capital stock. Figure 1 illustrates this concavity in the imbalance variable.

4.2 Fixed adjustment costs and stochastic adjustment costs

When a firm adjusts its capital stock, it pays a fixed cost proportional to foregone profits. The adjustment cost takes the form

$$\omega e^{\beta z^-} K^*,$$

where z^- is the pre-adjustment imbalance and ω is a random adjustment-cost draw. The key departure from the standard (S, s) model is that ω is drawn from a distribution $G(\omega)$ and varies across firms and over time.

This stochastic fixed-cost assumption is economically useful because actual firms do not always face the same cost of reorganizing, replacing capital, finding buyers for old machines, or implementing new investment projects. The same firm can face different costs at different moments, and different firms can face different costs at the same imbalance. This transforms the deterministic trigger rule into a probabilistic adjustment rule.

4.3 Firm adjustment rule

The firm compares the value of not adjusting with the value of adjusting to a return point c. For each imbalance z, there is a maximum adjustment cost $\Omega(z)$ such that the firm adjusts if $\omega \leq \Omega(z)$. Since adjustment is least attractive near the return point c, $\Omega(c) = 0$; as z moves away from c, $\Omega(z)$ rises.

The adjustment hazard is therefore

$$A(x) = G(\Omega(x + c)), \qquad x = z - c.$$

It gives the probability that a firm with imbalance x adjusts. The hazard is increasing in $|x|$: firms with larger shortages or excesses of capital are more likely to undertake a lumpy adjustment.

5 Aggregation and aggregate investment equation

5.1 Cross-sectional distribution of imbalances

Let $f(x, t)$ denote the cross-sectional density of firm-level imbalances before adjustment in period t. Since firms have heterogeneous histories of shocks and adjustment decisions, this density is generally nondegenerate. Aggregate investment is obtained by integrating expected firm-level adjustment across this density.

If a firm with imbalance x adjusts, it moves back to the return point and invests approximately $(e^{-x} - 1)$ times its capital. Multiplying this adjustment size by the probability of adjustment yields expected investment conditional on x. The approximate aggregate investment-capital ratio is therefore

$$\frac{I_t^A}{K_t^A} \approx \int (e^{-x} - 1) A(x) f(x, t)\, dx.$$

The nonlinearity is immediate: because $(e^{-x} - 1)A(x)$ is nonlinear in x, aggregate investment depends on higher moments of the cross-sectional distribution, not just the mean disequilibrium.

5.2 How the model nests linear partial adjustment

The standard partial-adjustment model appears as a limiting case. If the adjustment hazard is constant and the distribution of imbalances is concentrated near zero, then $e^{-x} - 1 \approx -x$ and

$$\frac{I_t^A}{K_t^A} \approx -A_0 \bar{x}_t,$$

which is the linear partial-adjustment representation. Thus, the paper does not simply reject linear models by assumption; rather, it embeds them as a special case and estimates whether the data prefer increasing hazards and nonlinear dynamics.

6 Empirical strategy

6.1 Data

The empirical analysis uses annual gross investment and capital series for 21 two-digit U.S. manufacturing industries from 1947 to 1992. The authors analyze equipment and structures separately, yielding two panels of investment-capital ratios beginning in 1948. The capital stock series are the Bureau of Labor Statistics series used in productivity studies.

6.2 Parameters and likelihood

The econometric problem is to estimate the adjustment-cost/hazard parameters while treating other structural components parsimoniously. The main source of randomness in the likelihood is the sectoral shock. Conditional on an initial cross-sectional density and hazard parameters, the model maps a sequence of sectoral shocks into observed sectoral investment rates. Because the mapping from the current aggregate shock to investment is one-to-one, the authors can recover shocks implied by observed investment and evaluate their likelihood.

6.3 Two estimated specifications

The paper estimates two versions of the model.

1. **Semi-structural hazard model.** The adjustment hazard is directly parameterized as an inverted normal:
$$A(x) = 1 - e^{-A_0 - A_2 x^2}.$$
The key hypothesis is whether $A_2 > 0$, meaning that adjustment probability rises with the size of imbalance.

2. **Structural model.** The distribution of adjustment costs is assumed to be Gamma, and the hazard is derived by solving the firm's dynamic optimization problem. The estimated parameters are the mean and coefficient of variation of adjustment costs.

7 Main empirical findings

7.1 Increasing adjustment hazards

The semi-structural estimates reject the constant-hazard model in favor of an increasing hazard for both equipment and structures. This means the probability of adjustment is strongly state dependent: firms with larger capital imbalances are much more likely to adjust. For equipment, the estimated hazard rises from roughly 14 percent for small imbalances to about 45 percent for a 40 percent imbalance. For structures, the hazard is near zero for small imbalances and rises to roughly 32 percent for a 40 percent imbalance.

7.2 Structural adjustment costs

The structural estimates imply substantial fixed adjustment costs. The estimated average adjustment-cost draw equals about 16.7 percent of annual operating profits for equipment and 22.8 percent for structures. Because firms can wait for favorable draws, the average cost actually paid conditional on adjustment is lower, especially for equipment. Structures appear to have larger and less variable adjustment costs, consistent with the intuition that structure investment is more irreversible and harder to reorganize than equipment investment.

7.3 Fit relative to no-adjustment benchmark

The likelihood gains relative to a no-adjustment/no-dynamics benchmark are large. This supports the view that adjustment costs and dynamics are empirically central for explaining sectoral investment. The structural model improves especially for structures, where lumpy and fixed-cost features are likely to be more pronounced.

7.4 Comparison with linear alternatives

The paper compares the nonlinear structural model with standard linear alternatives: a partial-adjustment/AR(1)-type model and a more flexible AR(2). The nonlinear model has higher likelihoods and generally superior nonnested-model test performance. Forecasting tests also favor the nonlinear model: the model improves one-step-ahead forecast mean squared errors in a large fraction of sectors, especially at horizons where accumulated cross-sectional imbalances matter.

7.5 Dynamic implication: nonlinear amplification after accumulated shocks

A key dynamic implication is that investment responds more strongly when the cross-sectional distribution has accumulated many firms near adjustment regions. In normal times, many firms remain passive. But after a sequence of shocks has pushed many firms toward sizable imbalances, the same aggregate shock can trigger a larger burst of adjustment. The model therefore produces periods of brisk aggregate investment response without imposing ad hoc time variation in parameters.

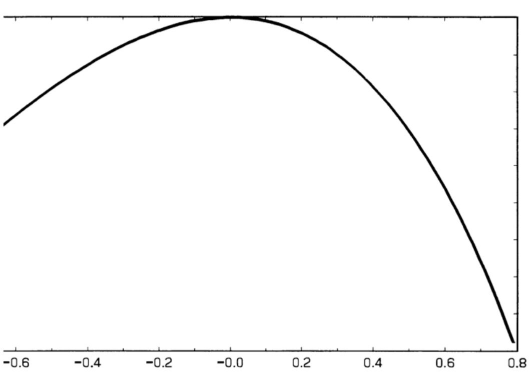
8 Identification and interpretation

The paper's identification comes from the nonlinear serial dependence in sectoral investment rates. A linear model can match average persistence, but it cannot easily match changes in persistence and responsiveness over time that arise from the evolving distribution of imbalances. In the generalized (S, s) model, lagged aggregate investment contains information about the cross-sectional density, and the density determines the current elasticity of aggregate investment to shocks.

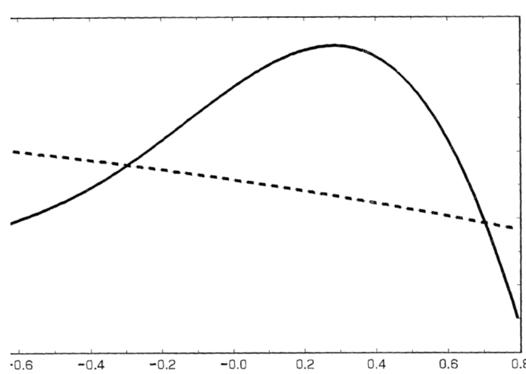
The model's structural interpretation should be read carefully. The authors do not observe plant-level adjustment costs directly in the estimation. Instead, they infer adjustment-cost parameters from aggregate sectoral investment dynamics, under assumptions about shocks, depreciation, and the distribution of idiosyncratic shocks. The value of the exercise is that the estimated nonlinearities have a coherent microeconomic interpretation and improve empirical performance.

9 Figures and tables extracted from the paper

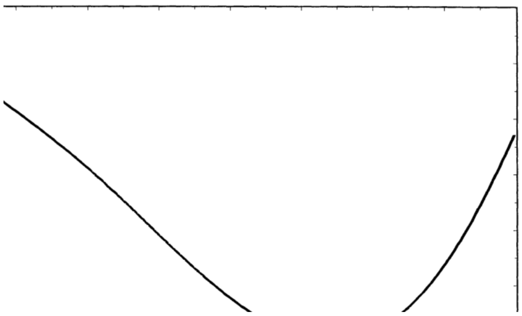
The following original figures and tables are extracted from the paper. Paired items are placed two per row; where a row has a single leftover item, it is displayed at width 0.6.



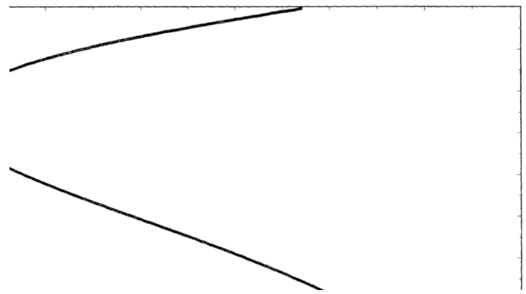
(a) **Figure 1**. Profit per unit of frictionless capital, showing how profits fall as the capital imbalance moves away from the target.



(b) **Figure 2**. Value of not adjusting and value of adjusting for a given cost draw; intersections define the inaction region.



(a) **Figure 3A**. The threshold function $\Omega(z)$, the maximum adjustment cost that still induces adjustment.



(b) **Figure 3B**. The implied lower and upper trigger bands as functions of the adjustment-cost draw.

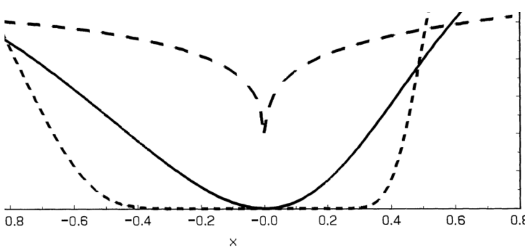


FIGURE 4A

(a) **Figure 4A**. Adjustment hazards under alternative distributions of stochastic adjustment costs.

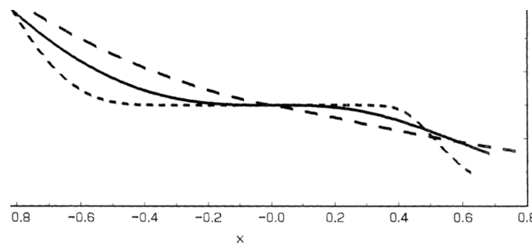


FIGURE 4B

ature not shared by the standard quadratic adjustment cost model but it is certainly
models with fixed, or even nonconvex, adjustment costs.
nanufacturing plant level data studied in Caballero, Engel, and Haltiwanger (1995)
e average observed distribution of disequilibria is considerably more platokurtic than
ribution of shocks affecting these plants.

(b) **Figure 4B**. Expected investment as a function of imbalance, combining adjustment probability and adjustment size.

ıoı ıne average costs enecuvely paıu by nrms wnen going unougn
:ment episode. Indeed, the average costs *paid* are 11.1 percent for

TABLE I

MAIN RESULTS

:rs	Equipment		Structures	
	Semi-structural	Structural	Semi-structural	Structural
	0.155		0.000	
	(0.067)		(0.054)	
	2.804		2.437	
	(1.165)		(0.878)	
	0.057	−0.006	0.013	0.019
	(0.013)	(0.016)	(0.006)	(0.002)
		0.166		0.228
		(0.029)		(0.046)
		0.327		0.066
		(0.109)		(0.009)
	2430.2	2431.4	2612.4	2637.2
.DJ	2315.2	2315.2	2497.0	2497.0

:andard deviations in parenthesis. LLK: log-likelihood. LLK-NADJ: log-likelihood without
: costs/dynamics.

(a) **Table I**. Main semi-structural and structural estimates for equipment and structures.

$$= \frac{}{\sqrt{\hat{S}T}},$$

l Normal distribution. Positive values of $V_{1,2}$ indicate evidence in
l 1; negative values evidence in favor of model 2.[28]
esents Vuong's statistics and the *p*-values for the test that both
and nonlinear) are equally close to the "true" model, against the
ıt the (nonlinear) structural model is closer. It is apparent that the
s is rejected in favor of the alternative even at very low signifi-

TABLE II

NONNESTED MODELS TEST: NORMAL SHOCKS

	Equipment		Structures	
	PAM	AR(2)-UNC	PAM	AR(2)-UNC
statistic	2.61	4.25	2.92	4.07
	[illegible]	[illegible]	[illegible]	[illegible]

(b) **Table II**. Vuong nonnested tests comparing nonlinear structural models with linear AR alternatives under normal shocks.

ɔg-Normal distributions. The log-Normal component does not add
ı the structural model, while the linear models assign most of the
log-Normal component. For this reason we compare the structural
Normal shocks versus the linear models with log-Normal shocks in
hough the likelihood in the linear model improves substantially
ification, Table IV shows that the test for nonnested models still
nlinear) structural models by a wide margin. In fact, the reduction

TABLE III

SKEWNESS AND KURTOSIS FOR INNOVATIONS

	Equipment		Structures	
Model	Skewness	Kurtosis	Skewness	Kurtosis
art. Adj.	0.49	1.15	0.95	1.88
	(0.13)	(0.36)	(0.15)	(0.65)
.R(2)	0.38	1.00	0.86	1.65
	(0.12)	(0.31)	(0.14)	(0.60)
tructural	−0.04	0.65	0.17	0.79
	(0.11)	(0.20)	(0.13)	(0.34)

(a) **Table III**. Skewness and kurtosis of innovations implied by alternative models.

TABLE IV

NONNESTED MODELS TEST: LOG-NORMAL / NORMAL SHOCKS

	Equipment		Structures	
	PAM	AR(2)-UNC	PAM	AR(2)-UNC
ormal)	2.42	7.97	3.57	4.53
:	0.0078	< 0.0001	0.0002	< 0.0001
k.	2409.3	2453.8	2611.3	2678.6

Vuong statistic calculated as in (20). All test statistics compare the structural model
d in Table I with the model in the table. If both models are "equally good," the
tic distribution of the statistic is Standard Normal. Large positive values provide evi-
ı favor of the structural model.

inator due to the increased precision of the test (the likelihoods
correlated across models) more than outweighs the increase in the
he linear model.

Criteria

(b) **Table IV**. Nonnested tests using a log-normal/normal shock specification.

.(2)	MODEL	ln((1)/2))	AR(2)	MODEL	ln((1)/(2))
;91	0.640	−0.075	0.306	0.340	−0.092
:40	0.230	0.038	0.371	0.220	0.502
;08	0.270	0.131	0.207	0.150	0.359
;66	0.540	0.044	0.507	0.350	0.371
;99	0.330	0.194	0.518	0.340	0.425

ırameters were estimated using data up to 1987. MSEs are multiplied by 10^3.

(1996) for arguments on the bias against nonlinear models inherent in MSE

(a) **Table V**. Sectoral average mean squared errors for one-step-ahead forecasts.

	0.078	0.076	0.023	0.017	0.024	−0.352
	1.739	1.410	0.211	0.201	0.270	−0.288
	0.551	0.730	−0.276	0.353	0.400	−0.127
	0.506	0.420	0.181	0.328	0.390	−0.170
	0.310	0.220	0.338	0.099	0.150	−0.411
	0.241	0.270	−0.124	0.494	0.075	1.888
	0.221	0.200	0.120	0.184	0.210	−0.124
	0.099	0.072	0.319	0.162	0.100	0.467
,(2))	—	—	0.047	—	—	0.312
	0.241	0.220	0.003	0.328	0.210	0.159

arameters were estimated using data up to 1987. MSEs are multiplied by 10^3.

(b) **Table VI**. Fraction of sectors for which nonlinear forecasts beat the AR(2) benchmark across horizons.

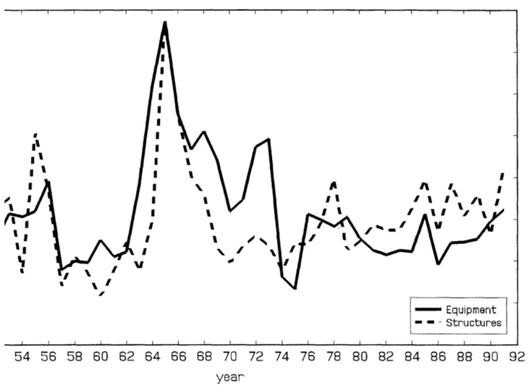


FIGURE 5

(a) **Figure 5**. Median one-step-ahead forecast paths for equipment and structures.

(b) **Table VII**. Average investment conditional on the sorting variable that captures the distributional state.

TABLE VIII

ASSESSING THE APPROXIMATION FOR AGGREGATE INVESTMENT

	Equipment	Structures
$\hat{a}$	0.015	0.003
$\hat{b}$	1.054	1.080
R_1^2	0.980	0.995
$\hat{a}$	-0.019	-0.005
$\hat{b}$	1.225	1.145
R^2	0.996	0.999

2. ADJUSTMENT HAZARD

we study the properties of the adjustment hazard:

$= G(\Omega(x+c)),$

l element (sav the smallest one) in the set $\mathscr{C}$ characterized in Proposition A5.

Table VIII. Monte Carlo evidence assessing the approximation used for aggregate investment.

10 Detailed interpretation of original figures and tables

10.1 Figures 1–4: from firm problem to adjustment hazard

Figures 1–4 provide the paper's model logic visually. Figure 1 shows that the firm loses profits when its capital stock deviates from the frictionless target. Figure 2 shows how adjustment is chosen by comparing the value of inaction to the value of paying the adjustment cost and returning to the target. Figures 3A and 3B translate this into a threshold rule: for each imbalance, there is a maximum cost draw for which adjustment occurs, and equivalently a pair of trigger points for each realized cost. Figures 4A and 4B are the most important for aggregation: once the cost draw is integrated out, deterministic triggers become a smooth hazard, and expected investment becomes a nonlinear function of imbalance.

10.2 Table I: main parameter estimates

Table I is the key empirical table. It shows that the increasing hazard parameter is significant in the semi-structural specification for both equipment and structures. In the structural model, the estimated mean adjustment costs are economically large, with structures having a higher mean and a much lower coefficient of variation. The likelihoods are much higher than the no-adjustment benchmark, indicating that the model's dynamic adjustment mechanism is important empirically.

10.3 Tables II–IV: linear versus nonlinear dynamics

Tables II–IV compare the nonlinear structural model with linear alternatives. The normal-shock Vuong tests favor the nonlinear model, while the log-normal robustness exercise addresses the concern that the nonlinear model's advantage might come from non-normal residuals rather than structural nonlinearities. Even when the linear model is allowed a more flexible shock distribution, the nonlinear model generally remains competitive or superior, especially for structures.

10.4 Tables V–VI and Figure 5: forecasting performance

Tables V and VI evaluate out-of-sample forecasting. The nonlinear model's advantage is not only in fitting the sample but also in predicting sectoral investment. The forecasting gain is particularly informative because the model's nonlinear state variable - the evolving cross-sectional density - is designed to matter for future responsiveness. Figure 5 illustrates the resulting forecast dynamics: nonlinear and linear forecasts can diverge when the model infers that the sector is in a high-responsiveness state.

10.5 Tables VII–VIII: mechanism and approximation

Table VII connects the estimated model to its mechanism by showing that average investment differs depending on the sorting variable related to the distribution of imbalances. Table VIII checks that the approximation used to express aggregate investment as a function of the cross-sectional density is accurate enough for the empirical exercise. This is important because the tractability of the model relies on an aggregation approximation.

11 How to position this paper in a literature review

11.1 Relation to investment-Q and partial-adjustment models

Traditional empirical investment equations often rely on smooth convex adjustment costs or partial adjustment. In those models, investment responds gradually and linearly to fundamentals. Caballero and Engel argue that this misses the micro-level lumpiness documented in plant data and the resulting nonlinear aggregation. The paper can therefore be positioned as a bridge from micro evidence on lumpy investment to aggregate investment dynamics.

11.2 Relation to (S, s) models

Standard (S, s) models have sharp inaction bands and deterministic triggers. They are conceptually powerful but difficult to aggregate and sometimes too stark empirically. Caballero and Engel's stochastic fixed-cost formulation preserves the lumpy adjustment logic while producing smooth hazards. This makes the model empirically tractable and flexible.

11.3 Relation to heterogeneous-agent macro and aggregation

The paper is an early example of how aggregate dynamics can depend on the distribution of micro states. This idea is now central in heterogeneous-agent macroeconomics. The paper's cross-sectional density of imbalances plays the role of a sufficient state variable for aggregate investment dynamics, analogous to how wealth, productivity, or balance-sheet distributions matter in later heterogeneous-agent models.

12 Strengths

- **Micro-founded nonlinear aggregation.** The paper derives aggregate nonlinearities from firm-level lumpy adjustment rather than imposing them statistically.
- **Flexible hazard formulation.** Random adjustment costs generate smooth hazards that nest both sharp (S, s) and approximately linear models.
- **Empirical discipline.** The model is estimated on sectoral investment data and tested against linear alternatives using likelihood and forecasting criteria.
- **Clear mechanism.** The mechanism - changing mass of firms near adjustment regions - gives a concrete explanation for time-varying aggregate investment responsiveness.
- **Separate equipment and structures analysis.** The paper shows that the mechanism is present for both categories but stronger for structures, matching economic intuition about adjustment costs.

13 Limitations and caveats

- **Aggregate estimation of micro frictions.** The adjustment-cost parameters are inferred from aggregate sectoral dynamics rather than directly estimated from plant-level cost data.
- **Distributional assumptions.** The likelihood depends on assumptions about sectoral shocks, idiosyncratic shocks, and the adjustment-cost distribution.
- **Approximation in aggregation.** The aggregate investment equation uses an approximation that abstracts from some correlation between firm capital size and imbalance.
- **Common hazard across sectors.** For tractability, the hazard or adjustment-cost distribution is imposed common across sectors, even though sector-specific technologies may differ.
- **Annual sectoral data.** Annual two-digit data may hide richer higher-frequency or plant-level heterogeneity.

14 Useful takeaways for future research

This paper is especially useful for research projects where micro-level adjustment is discrete but aggregate outcomes are smooth only because of aggregation. The logic can be adapted to investment, employment, inventories, price adjustment, durable purchases, credit adjustment, and technology adoption. The broader lesson is that aggregate dynamics may appear sluggish in normal times but become highly responsive after sufficient micro-level imbalances accumulate.

For empirical work, the paper suggests that lagged aggregate outcomes can encode information about hidden cross-sectional distributions. For theory work, it shows that adding heterogeneity in adjustment thresholds can make nonconvex adjustment models much more empirically tractable. For policy analysis, it implies that the effect of an aggregate shock depends on the economy's distributional state, not only on the shock's size.

15 Potential connection to firm dynamics, trade, and tariff research

The generalized (S, s) logic is relevant for studying firm responses to policy wedges such as tariffs. If firms face fixed costs of changing suppliers, relocating production, reorganizing capital, or adjusting product lines, then the response to tariffs will be lumpy and state dependent. A tariff shock may have a small short-run aggregate effect when few firms are near adjustment thresholds, but a large effect after accumulated exposure pushes many firms close to reoptimization. Similarly, exemptions, reversals, or temporary tariff suspensions may have asymmetric effects if firms have already paid adjustment costs or moved to new supply-chain positions.

This paper therefore provides a useful conceptual template for dynamic tariff projects: replace the capital imbalance with a policy-exposure or sourcing-gap state variable, replace investment adjustment with sourcing/capacity/product-line adjustment, and study how aggregate responses depend on the distribution of firm-level exposure gaps.

16 Key equations to remember

1. **Capital imbalance:**
$$z = \ln(K/K^*), \qquad x = z - c.$$
2. **Adjustment threshold:**
$$\Omega(z) = e^{(1-\beta)z}\{v(c) - v(z)\}.$$
3. **Adjustment hazard:**
$$A(x) = G(\Omega(x + c)).$$
4. **Expected firm-level investment:**
$$E[I_t(x)|x] = A(x)(e^{-x} - 1)K_t(x).$$
5. **Approximate aggregate investment-capital ratio:**
$$\frac{I_t^A}{K_t^A} \approx \int (e^{-x} - 1)A(x)f(x,t)\,dx.$$
6. **Linear partial-adjustment special case:**
$$\frac{I_t^A}{K_t^A} \approx -A_0\bar{x}_t.$$

17 Bottom line

Caballero and Engel show that aggregate investment dynamics are better understood as the aggregation of many lumpy micro-level adjustments than as a smooth representative-firm process. Random fixed adjustment costs generate smooth adjustment hazards, and the evolving cross-sectional distribution of imbalances creates nonlinear aggregate investment dynamics. Empirically, this framework fits and forecasts U.S. manufacturing equipment and structures investment better than standard linear models, especially because it captures how the number of firms ready to adjust changes over time.